

L4

Intermediate loops

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Type Casting

Implicit → Compiler will convert automatically

int a = 4;

double b = a;

4.0

int c = 4;

long d = c;



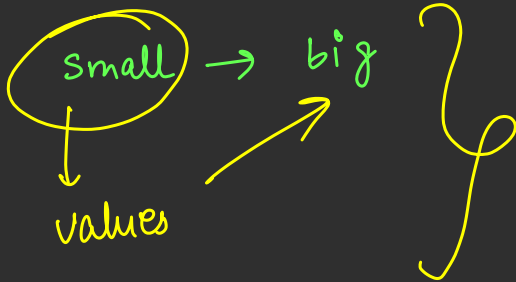
↓
long

↙ int

long

int → -2^*10^9 to 2^*10^9 (smaller)

long $\rightarrow -2 * 10^{18}$ to $2 * 10^{18}$ (big)

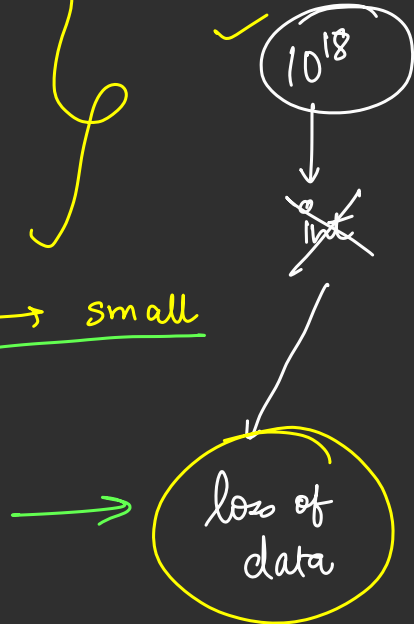


long a = 4;

int b = (int) a;

Force Convert

big $\rightarrow$ small



big → small



loss of data

Explicit type
casting

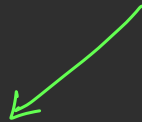


```
double c = 8.23;
```

```
float d = c;
```



Error



(float)

```
float d = (float) c;
```

Type casting `int a = 1000;`
 | → integer

```
long b = 1000;
```

integer

long b = 1000 000 000 000 L / l

double a = 5.0 ;

└─→

Decimal ~ double ;

float b = 4.5 ;

big to small

float b = (float) 4.5 ;

↓

float

└─→ decimal

float b = 4.5 f ;

└─→ float }

while $\rightarrow$ digits

for $\rightarrow$ start - - - - - end

for (initialization ; test condition ; update expression)

for (int i = 1; i <= 3; i = i + 1;)
{
 S.O. Pln (i);
}

① $\rightarrow$ initialization
② $\rightarrow$ test condition
③ $\rightarrow$ update expression
④ $\rightarrow$ loop body

i = 4, 4 <= 3

$\downarrow$
false $\rightarrow$ STOP

1
2
3

~~i = 1~~
~~2~~
~~3~~
4

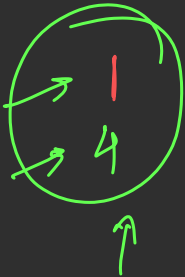
```
for (int i = 1; i <= 5; i = i + 3)
```

{

S.O.Plm(i);

}

$i = i + 3$
 $i = 4$
 $i = 7$



Step I

$i = 1, 1 \leq 5, \text{ print } 1, i = 4$

Step II

$i = 4, 4 \leq 5, \text{ print } 4, i = 7$

III

$i = 7, 7 \leq 5$
 $\rightarrow \text{false, STOP}$



Print all numbers from 1 to n .

↳ User input

$n \rightarrow$ input

1 to $n \rightarrow$ for loop



Print all odd numbers from 1 to n

while $\rightarrow$ digits

1 2 3 4



4, 3, 2, 1

Initialization

while (test condition)

{

update statement

}

int i = 1; → ①
while (i <= 3) → ②
{
 S.O.Pln (i);
 i = i + 1; → ④

}

I, i = 1, 1 <= 3, i = i + 1 = 2

II, i = 2, 2 <= 3, i = i + 1 = 3

III, i = 3, 3 <= 3, i = i + 1 = 4 → stop

1
2
3

457

$$\lfloor n/10 \rfloor \rightarrow \frac{457}{10} = \underline{\underline{45}}$$

Find last Digit $\rightarrow$ $n \% 10$ = last digit

Remove last digit



$$n = n/10;$$

$$n = n/10$$

$$\begin{array}{r} 10 \overline{) 457} \\ \underline{450} \\ 7 \end{array}$$

→ Remainder

$$n = 457$$

7 5 4

reverse digits ↑

```
while ( n > 0 )
```

```
{
```

```
    int lastDigit = n % 10;
```

```
    S.O.Plus (lastDigit);
```

```
    n = n / 10;
```

```
}
```

LD = 7 7

n = 45

LD = 5 5

n = 4

LD = 4 4

n = 4